# The Open Source Firefighter

Annex A — Reference Hardware Specification

**v1.0 · August 2026 · Reference hardware · Companion to the Main Specification** *Document ID:* ***OSF-1.2*** *· one document of the Open Source Firefighter specification family.*

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*This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.*

*They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.*

*All external standards cited in this annex are adopted by reference to their current edition as amended — the issuing authority's text governs; see the specification's Adoption by Reference standing rule.*

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## Scope

This annex defines reference hardware for compliant installations. It specifies capability requirements first and names example products second. Substitution of any component meeting the capability requirements is expressly permitted and expected. Every component class is selected for availability through standard industrial supply channels. No component in a compliant installation may depend on a single manufacturer for replacement.

**Three enclosure classes, one system.** The hardware family consists of three enclosure classes — **Residential**, **Government/Commercial**, and **Tactical** — running identical firmware, identical coordination protocol, and identical message categories. The classes differ in packaging, connectors, power sourcing, and environmental rating only. Class boundaries are environmental exposure and procurement requirements, never capability. A government installation is not a smarter unit; it is the same unit in harder packaging.

**Expansions are orthogonal to class.** Pump and reservoir expansion, external zone modules, the gateway kit, and detection payloads attach to any class where noted. Capability grows by module; the core unit does not change.

**Equipment category and certification pathway.** This equipment is a supplemental exterior wetting system for ember defense: an industrial control panel operating irrigation-class hydraulics. It is not a fire alarm system (NFPA 72 / UL 864 class), not a fire suppression system as defined by the sprinkler codes (NFPA 13, 13D, and 13R), and not a life-safety notification device, and shall not be represented, marketed, or certified as any of these. Where a jurisdiction regulates construction and mitigation at the wildland-urban interface — commonly through the International Wildland-Urban Interface Code, NFPA 1140, or NFPA 1144, and in some states through a dedicated chapter such as California Building Code Chapter 7A — this equipment is supplemental to those requirements and replaces none of them. Compliant units are certifiable as industrial control panels under UL 508A-class standards through the existing multi-vendor panel-shop ecosystem; component-level recognition is carried by the commodity bill of materials. This specification's equipment class does not invoke fire-alarm control unit listing regimes (UL 864 class), and no revision shall move it into one.

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## Design Principles

**Sequencing over splitting.** A single supply point — a hose bib, a pump, a fuel can — is time-shared across coverage points rather than divided among them. Default operation opens one zone at a time on a rotating schedule. This is the correct hydraulic design for the supply that exists, and the correct fire physics: a wetted surface remains protective for minutes between passes, and duty-cycled coverage defeats ember ignition at a fraction of the water and fuel. Sequencing at the unit level is the same algorithm the mesh runs at neighbourhood level and the domain runs at valley level.

**Commodity wear items.** Every component with a service life — batteries, valves, fuses, sensors — shall be a commodity item replaceable by any competent person with a supply house account and no documentation beyond the component label. Battery charging and DC bus management shall be provided by standard DIN rail power modules. Integration of charging circuitry onto proprietary control hardware is expressly non-compliant.

**One control object at every size.** Zone valves are quarter-turn valve bodies with ISO 5211 actuator mounting pads

driven by 24VDC electric actuators. A 3/4” brass ball valve and an 8” lug butterfly valve present identical control interfaces: open, close, limit-switch position feedback. Scaling changes valve body and actuator torque class only.

**Minimum viable installation.** One garden hose, one sprinkler, one zone. The Residential unit shall provide scheduled surface wetting and full mesh participation with a single consumer-grade sprinkler on garden hose at 40 psi municipal supply. Two to three sprinklers across the three provided zones shall achieve meaningful roof and perimeter coverage for a single-wide manufactured home, tiny home, cabin, or small residence.

**No novel protocols where standard ones exist.** Routing, gateway selection, addressing, and power delivery negotiation use standard mechanisms with production histories. This specification invents only at the coordination layer.

**Operators prevail.** Automation in this system manages resources; it does not overrule operators. Where automated policy and operator action conflict at the point of use, the operator prevails and the system’s role is to make consequences visible. (Main specification.)

**Designed for the public.** The system is designed to be operated, moved, and understood by untrained members of the public; familiarity, not certification, is the intended long-term relationship between communities and this equipment. Voice guidance, hardware controls, one-person weight limits, and the absence of any account, application, or literacy barrier all serve this principle. The long-term benchmark is the fire hydrant: universally recognized, universally trusted, understood without training.

**The filter and the floor converge.** The design filter is the hardest user — the expert who will overload, improvise with, stand on, and defeat any equipment given to them. The design floor is the newest user — the untrained volunteer encountering the equipment for the first time. These requirements converge: equipment that survives expert contempt and equipment that welcomes inexperience are the same equipment — simple, verbal, tactile, forgiving, heavy-handled, with nothing delicate, nothing clever, and nothing that requires reading. Where a design choice appears to serve only one of these users, it has probably failed both.

**Novelty is spent only where the invention lives.** The coordination layer is new; everything the eye and hands touch shall be the most familiar possible object. The unit deliberately resembles a fire alarm panel at the component level — NEMA enclosure, 24 VDC bus, DIN internals, shelf-mounted commodity batteries, status LEDs, audible annunciation, a maintenance chirp — because resemblance is inherited trust: inspectors approve what they recognize, the alarm and controls trades already know the work with their hands, and the maintenance habits society runs for panels and smoke detectors transfer whole. The enclosure earns trust by familiarity so the invention inside never has to spend any.

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## Common Architecture (All Classes)

The following internals are common to every enclosure class.

### Controller

- ESP32-class industrial PLC, DIN rail, 24VDC (Industrial Shields ESP32 PLC 21/38 series, Norvi ENET series, or equivalent)
- Minimum I/O: 8 digital outputs (valve drive, pump-start relay, status), 8 digital inputs (valve limit switches, flow pulse, GNSS PPS), 2 analog 4–20mA inputs (pressure, expansion), 1 analog input reserved for acoustic sensing where equipped
- IEC 61131-3 environment or open firmware toolchain (ESP-IDF / Arduino class)
- Serial or Ethernet interface to radio and router function
- Router function (publication policy, message signing, protocol handling) runs as firmware on the controller or radio MCU at Residential class; dedicated router hardware at Government/Commercial and Tactical class where MANET-class integration or publication policy complexity requires it

### Zone Valves and Manifold

- Three standalone motorized 3/4” full-port two-piece brass ball valves, 24VDC, ISO 5211 actuator pads, position feedback (CR-series class), fed by a basic 3/4” intake manifold header; inlet enters the wet section side wall, outlets exit the enclosure bottom, one per zone. Valves mount individually — replacement is one valve, not an assembly. (Solenoid valves are a permitted budget variant where position feedback is deliberately waived; the reference design requires feedback.)
- Common inlet gallery with spring check valve and manual isolation valve

**Backflow — hazard class and what the check valve is not.** The inlet gallery check valve is a system component;

it is **not the backflow prevention device** and shall not be represented as one. Backflow protection is a separate, listed assembly selected and installed to the local plumbing code by the authority having jurisdiction. Note the hazard class: an installation that can draft from a pond, cistern, or other non-potable source through the same manifold that also connects to a potable supply is a **high-hazard cross-connection**, which typically requires a reduced-pressure-zone assembly or a physical air gap rather than a simple check or dual-check device. Where stored and potable sources share plumbing, the air gap is the conservative answer. - Drain-down port at manifold low point for winterization (firmware-assisted routine: all valves open, supply isolated, drained, depressurized) - One 0–200 psi pressure transducer, 4–20mA, on the inlet gallery - All wetted connections NPT; GHT adapters supplied for inlet and each outlet - **Self-calibration:** the controller opens each zone individually at commissioning and periodically thereafter, observes inlet pressure sag, and sets cycle timing automatically. No hydraulic configuration by the user. - **Sequencing default:** one zone open at a time, rotating. Simultaneous operation is not a requirement and shall not be assumed. - Expansion beyond three zones: external zone modules, same valve family, small NEMA enclosure, daisy-chained 4-wire bus (24VDC pair, signal pair)

### Radio and Network

- LoRa mesh node, Meshtastic protocol or compatible, operating in the jurisdiction’s unlicensed ISM allocation (902–928 MHz at 100 mW is the North American reference; use the regional allocation and power limit elsewhere), (RAK4631, T-Beam, Heltec class), external antenna via bulkhead connector — reference radio at Residential class
- **Unlicensed allocations only.** A compliant node operates under the unlicensed rules of its jurisdiction (Part 15 in the United States, or the local equivalent) and shall not be configured for amateur-band operation. Amateur allocations are outside this specification for several independent reasons: amateur operation requires a licensed control operator and station identification at intervals, which an unattended automatic node cannot satisfy; amateur rules restrict unattended automatic operation and bar communications that could reasonably be furnished through other radio services on a regular basis; and amateur allocations may not be used for communications in which the operator has a pecuniary interest. A signed, unattended, continuously operating node is not an amateur station. Firmware shall not offer an amateur-band mode, and a node with message signing disabled is non-compliant regardless of band.
- **Tactical MANET radio class** — Government/Commercial and Tactical option: agency-furnished mobile ad-hoc mesh data radios of the hardened interagency class (Silvus StreamCaster, Persistent Systems Wave Relay, TrellisWare TSM, and peers). The class is defined by capability — self-forming mesh, IP presentation, MIL-STD-810-class environmental qualification — never by vendor; any radio the furnishing agency fields and trusts is compliant at this layer.
- **Interagency governance, adopted by reference:** radio equipment operating on interagency wildland fire networks is governed by the National Interagency Incident Communications Division (NIICD) approved-equipment listings, and interoperability channel use by the National Interoperability Field Operations Guide (NIFOG). Nothing in this specification transmits on, contends for, or requires those networks; where agency-furnished radios are themselves NIICD-listed or incident-assigned, they operate under the furnishing agency’s authority and the incident communications plan, per the Spectrum Posture section. This specification’s coordination traffic rides its own unlicensed or agency-furnished data layers and never the fire voice nets.
- **Satellite backhaul (optional, all classes):** commercial LEO (Starlink class, including compact low-power terminals) or government satellite transport (Starshield class or successors), presenting standard IP to the network layer. Backhaul is an enhancement, never a dependency.
- **Gateway node role:** any node may bridge the coordination domain to a backhaul. Gateway is a role, not a model. Gateways advertise external reachability via standard routing protocol mechanisms; nodes select per normal route metrics; failed gateways age out. This specification defines no novel gateway election protocol. Zero gateways is fully operational.
- **Export-control boundary:** this specification defines open interfaces at the network layer; export-controlled radio and transport equipment (tactical MANET class, government satellite transport) integrates as operator-furnished equipment through those interfaces. No controlled technical data is incorporated into this specification or its reference implementations, and the open project ships no controlled articles. Government configurations are assembled by the operator holding the equipment and the authorities.

**Spectrum posture.** Primary coordination communications shall use unlicensed spectrum under FCC Part 15 (or the local equivalent) or wired/commercial IP transport (Ethernet, copper, fiber, commercial satellite service); no compliant installation shall require a spectrum license to perform its fire defense function, and no compliant installation shall be made to require frequency coordination. The coordination protocol is designed to function correctly on interference-tolerant unlicensed spectrum: degraded delivery is a handled operating condition, not a failure mode. Licensed radio classes integrate solely as operator- or agency-furnished equipment under the license holder’s authority; agency-furnished radio classes operate under the furnishing agency’s spectrum management

(NTIA/FCC coordination, incident communications plans), and compliant equipment shall accept the operating parameters that authority assigns.

**Public safety trunked data (P25 class) — noted, not touched.** Existing public safety trunked radio systems include narrow-band data services technically sufficient for this specification's coordination traffic, and this specification's message sizes are compatible with them. Integration is architecturally supported through the router layer as an agency-furnished radio class and is left entirely to the owning agency's initiative. This specification does not contend for public safety spectrum or dispatch capacity.

- **Range-convergence doctrine.** The system's three operative ranges converge by physics, not coincidence: realistic terrain LoRa reach, practical hose-lay length, and interface distance (the structure spacing at which coordinated defense matters) all occupy the same neighborhood-scale envelope. The system coordinates best at exactly the distance where coordination matters most, over exactly the reach the hydraulics serve. Employment gradient: **dense interface** — full coordination, mesh multipath, shared supply intelligence (the design center); **scattered rural** — per-site autonomous clusters (a homestead is its own cell) linking opportunistically or by gateway; **true remote** — single-cluster autonomous with reporting reach via higher radio classes or backhaul, per the graceful-degradation architecture. Field rule of thumb: *if a hose could reach it, the mesh reaches it.*

### GNSS: Position and Time

Every node carries a GNSS receiver providing node position (system-status-class data; coarse-position-only publication is a supported policy option) and precision time (PPS-disciplined controller clock preferred) for the cryptographic log and coordination scheduling — the only precision time source consistent with Charter Clause 2. Degradation: no fix, fall back to RTC and last-known position; fire defense unaffected.

Capability: *GNSS receiver, serial or USB interface, NMEA output, PPS preferred*. Compliant antenna/receiver patterns:

1. **External active antenna, internal receiver (preferred):** commodity active puck through SMA bulkhead, top face; sky view required; co-location with LoRa antenna acceptable with a few inches of separation.
2. **External sealed puck on data cable:** weatherproof USB/serial puck (BU-353 class, marine NMEA) through a cord grip. Compliant; second choice.
3. **Integrated through-panel unit:** sealed GNSS whose mounting stub forms the penetration, per the assembled-penetration rule. Marine-grade hardware class; the standard configuration for wet, marine, coastal, mobile, and government-procured installations.

### Power System (Common Requirements)

- 24VDC bus throughout
- DIN rail supply with DC-UPS/charger function (Mean Well DRC-40/60 class, DR-UPS40 + supply, PULS UB equivalent); low-voltage disconnect required
- Battery: commodity form factor, field-replaceable — class-specific sourcing below
- Standby budget: controller, radio, GNSS draw single-digit watts; valves draw only during travel; 72-hour autonomous requirement met with margin at every class

### User Interface (Common Minimum)

- Panel face: status LEDs (power, radio, readiness, fault, active, distress)
- **Local operator interface (controller inputs, not software-only):** manual activation; disable/lockout; independent per-zone command; and distress notification by deliberate long-press. Required capability, not a fixed control count — the interface shall be operable in structural firefighting gloves, in darkness, without instruction, without a personal device, and without network or account. *Reference implementation: discrete panel pushbuttons with indicator lamps; a rotary selector, keyed switch set, or any other physical interface meeting the same capability is equally compliant.*
- Configuration via local-only service interface: the controller serves a plain web page over direct Ethernet or USB connection. No cloud, no app store, no account.
- Rich UI is the incident command interface (main specification); DIN-mount HMI panel is a permitted optional module

### Notification Button (All Classes)

The notification button is a controller input, not an emergency-stop and not a safety-rated device; it claims no safety

category. Actuation requires a deliberate press of roughly three to five seconds, which suppresses false triggers from brush contact or a bump. On a registered press the controller, at every class: publishes a distress event to the coordination domain (enumerated category: node identity, position, timestamp — deliberately minimal); latches the local system into full-protection mode (all-zone maximum cycle); and sets the node's status to distress, placing it on the incident command picture where one is operating. Per Charter Clause 7, no software configuration may disable the operator's controls; the actual hardware kill is the disconnect, and loss of power drives the auto-return valves closed independent of any control.

At Residential class, the button signals the coordination domain and anyone watching it; **it is not a substitute for emergency services, and this specification shall never claim otherwise.** At Tactical class, the same press additionally triggers the satellite emergency module where fitted (see Class 3). There is no reason to omit it from the Residential unit; the interface is identical, the tactical unit simply acts on it further.

### Audio Annunciation (All Classes)

Every unit carries a small speaker (Mylar cone or piezo element, weather-appropriate mounting or internal with sound port) driven by an I2S codec/amplifier module — commodity board-level parts native to the controller platform. The unit announces state in plain speech: system armed, activation started and ended, specific fault conditions ("zone two fault — valve did not confirm open"), low battery, supply pressure low, communications lost, maintenance required, and distress-transmitted confirmation on notification-button press. LEDs indicate that something is wrong; the voice says what.

- **Voice content is pre-rendered:** clips generated at firmware build time from a text manifest stored in the specification repository. No runtime synthesis dependency exists (Charter Clause 2); the same fault says the same words every time; and **the voice manifest is a specification artifact** — translations and localizations are community contributions like any other. On-device synthesis is permitted for composed values (readings, percentages) where clip permutation is impractical.
- **Annunciation is output only.** This section adds no microphone requirement at any class, and the Tactical acoustic engine sensor remains a machinery-state sensor per the message enumeration. Voice out; never audio in, on the mesh.
- **Maintenance discipline:** periodic audible maintenance reminder per the fire-panel tradition; interval and volume configurable, mute available in local configuration. Fault annunciation defaults on at Tactical class.

**Voice interface doctrine.** Voice follows the automated external defibrillator pattern: a closed, finite prompt set in plain imperative language, understandable by an untrained person under stress, requiring no training to operate. This is the design discipline codified across AED usability practice (IEC 62366 usability engineering, medical human-factors review) — proven at life-safety stakes on the general public. The prompt set is finite because the failure modes are finite; a few megabytes of pre-rendered audio covers the whole family. All classes share the same interface; the advanced tiers differ only by an extended clip set, not a different interface. Pre-rendered clips per the voice manifest (a specification artifact; translations are community contributions).

**Accessibility conformance.** Visual and audible indication are each other's required alternative mode, satisfying the Section 508 / Access Board ICT provision that any visual mode have a non-visual equivalent and any audible mode a visual equivalent. Every state announced by voice is also shown by indicator, and every indicator state is available audibly. A means to increase volume is provided. This is adopted by reference to the Section 508 ICT standards and the ADA audible-interface provisions (the talking-ATM and audible-elevator lineage); the Government/Commercial class is thereby accessibility-conformance-report (VPAT/ACR) ready for federal procurement, and the Residential class inherits conformance at no added cost.

**Audio is public; no private listening path required.** Audible status is public by design — the device announces only enumerated public status ("system armed," "zone two fault") and carries no private information through its audio channel. A standard audio output or coupler point is permitted where an installation wants one (high-noise environments, commercial preference) but is expressly not required: the Section 508 coupler provision protects private information, and a device with no private audio has nothing to couple. Public-access AEDs omit the jack for the same reason. Adopt the pattern; drop the vestige.

### Guided Setup (Tactical Class; Optional Elsewhere)

Tactical units commission by voice. Setup is a talking state machine — a fixed if-then checklist, not an intelligent system — in which each instruction gates on a sensor condition the controller verifies itself: "connect water supply to inlet" (advances on inlet pressure detected); "connect zone one" (advances on valve confirmation); "install antenna" (advances on mesh join or GNSS fix); "system ready — coverage active." The operator knows they are done because the unit says so.

Combined with the prepositioning requirements, this means a unit dropped to an operator who has never seen one commissions itself by voice, without a manual, in the dark, in gloves. Guided setup is a firmware feature, not additional hardware, and may be included at any class.

### Fasteners, Mounting, and Grounding

- **One interior fastener, one driver — best effort.** Interior mechanical fastening (rail to backplane, valve bracket, battery bracket, duct) standardizes on a single screw size and drive: M5 pan head, Torx T25 reference (Robertson square permitted). Where smaller hardware is unavoidable, the rule degrades to one driver, two bits, kept in the door pocket. Two finish sets: zinc-plated standard; 316 stainless for marine, coastal, and high-exposure builds (anti-seize on stainless-to-stainless — see Workmanship guide).
- **Mounting:** loaded wall-mount units run 45–60 lb wet. Four-point mounting into structure (studs or masonry anchors); drywall anchors shall not be used. External mounting feet or flanges shall be used so field-drilled enclosure penetrations are not required for mounting.
- **Height:** recommended installation places panel controls and indicators in the 48–60 in. AFF service band, which places the battery shelf at knee height (correct lifting) and the manifold drain at the bottom where gravity serves it. Site conditions govern.
- **Grounding and bonding:** a ground bar or bonding terminal shall be provided on the backplane; equipment grounding per the NEC with the supply circuit; antenna bulkheads (N, SMA) shall be bonded per established coax-entry grounding practice, adopted by reference. Electrical safety grounding (required, code-governed) is distinct from lightning protection (site-specific, outside this specification’s scope, addressed by existing standards).

### Wiring Practice (Reference Palettes)

Field-built and kit units are recommended to follow the established industrial palettes; single-color wiring with wire numbers and an internal legend is fully acceptable practice per the listing standard’s own terms. Color is the fastest documentation there is; the legend is the requirement.

- **Power (NFPA 79 convention, UL 508A-compatible):** blue — ungrounded 24VDC control (most of this panel); white/blue stripe — DC common; **orange — circuits that remain energized with the disconnect off**, which in this equipment is the battery/DC-UPS bus and its loops: this row is a safety marking, not an aesthetic; black — supply-voltage ungrounded (the 120VAC side); white — AC neutral; green or green/yellow — grounding. Factory sensor pigtails follow the IEC sensor convention (brown +, blue −, black signal) and are not re-colored.
- **Data (TIA-606 administration convention):** blue — horizontal runs to devices; purple — common equipment (controller, switch); green — network demarc/backhaul hand-off; yellow — auxiliary/alarm circuits (the event interface tap); fiber jacket codes adopted verbatim (yellow single-mode, aqua multimode). Labeling per TIA-606 administration is the underlying requirement at any color scheme.
- **Government tier — black-side doctrine:** all equipment under this specification, at every class, resides entirely on the unclassified (BLACK) side of agency red/black infrastructure separation. No RED-side cabling shall terminate on equipment defined by this specification; agency crypto and classified transport sit across that boundary under agency management, per the export-control section. Cable jacket color per agency convention makes this boundary visually inspectable at the open enclosure.

A companion **Workmanship & Best Practices guide** (recommendation tier: ferrules on stranded conductors, lever-terminal field wiring, dressing, labeling, service loops — “not required for compliance; expected in quality work”) is a queued specification-family artifact. The build layer that fire codes delegate to factories, this specification publishes, because its factory is everyone.

### Enclosure Water Management

Interior hydraulic/electrical cohabitation shall follow established sprinkler valve-enclosure and irrigation-controller practice, adopted by reference: control electronics mounted above the wet works and never beneath a mechanical joint; drip loops on all conductors descending toward the hydraulic zone; a drainage path from the enclosure low point to the exterior; and no battery placement below the hydraulic zone without a drained tray. Sectioned two-compartment enclosures or dual-enclosure (electrical + valve box) formats are equally compliant and may be preferred for first-time builds.

### Enclosure Penetrations — Assembled-Penetration Rule

Where a sealed component forms its own enclosure penetration (through-panel GNSS, combined antenna/radio pods, panel-mount connectors), the penetration is evaluated against the enclosure rating *as assembled*, not against the

reference fitting list. Reference layouts below are reference, not closed sets.

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## Class 1 — Residential

Wall-mounted unit for homes, manufactured homes, cabins, and small commercial structures on municipal or well supply.

- **Enclosure:** NEMA 4X (enclosure type ratings per NEMA 250 and UL 50E), hinged, lockable, nominal 16 × 20 × 8 in; polycarbonate or powder-coated steel (Hoffman, Rittal, Saginaw, Attabox class); battery shelf at bottom
- **Penetrations:** 3/4” NPT inlet (side); three 3/4” NPT outlets (bottom — the reference position, so the manifold gravity-drains; side outlets are permitted where the site requires, provided a drain path is maintained); N-type LoRa bulkhead (top); SMA GNSS bulkhead (top); cord grip (bottom/side)
- **Connectors:** commodity throughout — cord grips, DIN terminal blocks, NPT/GHT. Field-wireable with a screwdriver and strippers; nothing keyed or proprietary
- **Power:** 120VAC cord-and-plug; two 12V 7–9Ah SLA bricks in series (standard fire/security panel form factor) or drop-in 24V LiFePO4
- **Radio:** LoRa reference
- **Coverage assumption:** consumer-grade distribution — impact/oscillating sprinklers or irrigation heads on garden hose or 3/4” pipe; each zone drives one to four heads at 3–6 GPM; placement per main specification (roof line, gutters, ten-foot perimeter)

## Class 2 — Government / Commercial

Wall-mounted unit for federal facilities, commercial structures, critical infrastructure, and any installation procured against environmental-exposure or agency requirements. **Listing consideration:** a UL 508A panel shop generally cannot enclose an unlisted lithium pack; a lithium option should use cells and a battery system carrying UL 1642/UL 2054 (cell and pack) or UL 1973 (stationary battery system) listings, or the 508A route for the finished panel may be foreclosed. Sealed lead-acid is the routine case and carries no such constraint. Identical internals to Class 1; differences are packaging and options.

- **Enclosure:** NEMA 4X minimum; 316 stainless wetted fittings and polymer enclosures preferred for marine/coastal exposure
- **Connectors:** sealed circular connectors of an open multi-vendor standard for all external connections. **M12 reference class** (A-coded sensor/power, D/X-coded Ethernet); Deutsch DT for vehicle/rugged; MIL-DTL-38999 where procurement requires. Field units unplug and swap rather than rewire.
- **GNSS:** pattern 3 (integrated through-panel, marine grade) standard
- **Radio:** LoRa or tactical MANET tier; dedicated router hardware where publication policy or agency integration requires
- **Power:** hardwired or cord-and-plug per site; same DIN/commodity battery architecture
- **Internal networking:** DIN rail unmanaged industrial Ethernet switch, minimum 5 ports, 24VDC input from the existing bus, standard equipment (Phoenix Contact, Moxa, Antaira, Planet class). Port budget: controller, radio/router, backhaul terminal, external IP sensor or detection payload, service. External M12 X-coded Ethernet bulkheads terminate to the switch. **Unmanaged is the requirement; managed is permitted** where agency network policy requires it — the reference design shall never require network configuration to commission.
- **Blue force tracking:** capable of publishing node position and status as Cursor-on-Target messages to TAK infrastructure (ATAK/TAK Server) where deployed. Every node presents as a fixed friendly asset on crew situational awareness: a known point with water, power, communications, and a distress button.
- **Additional message categories:** thermal detection events, triage state, extended telemetry per the main specification’s tier enumeration
- **Options:** DIN-mount HMI, incident command integration, detection payload interface

### Maritime and Coastal Use Cases

The Class 2 marine-hardened configuration (316 stainless wetted fittings, sealed connectors, polymer enclosure, marine GNSS) already provisions the unit for salt-water environments; the maritime cases are that hardware pointed at new targets, with an effectively unlimited raw-water supply. The coordination primitive is unchanged. Representative cases, naming the category so coastal and maritime resources are covered by name:

- **Marina and mooring-field defense.** Packed hulls and fuel make a marina a continuous fuel bed with the worst possible structure spacing — vessels in contact spread fire hull-to-hull exactly as structures do below 30-foot separation, and raw water is immediately available. Dockside nodes drafting raw water to wet vessels and finger piers in rotation apply the sequencing thesis with the supply constraint removed.
- **Laid-up, reserve, and idle-fleet vessels.** Reduced-manning layup (reserve fleets, anchored idle tonnage, mothballed hulls) combines high value, low crew, and elevated ignition risk because no one is aboard to catch a start. A prepositioned node with detection and raw-water draft is the fire-lookout configuration afloat: passive until triggered, reporting by satellite, holding until a crew arrives — the Tactical prepositioning doctrine with a hull in place of a conex.
- **Ports, boatyards, and working waterfront.** Fuel docks, haul-out yards, and shoreside storage are the Class 2 unit performing its ordinary role near salt water.

**Regulatory boundary.** The maritime environment is governed by its own authorities — U.S. Coast Guard, SOLAS, classification societies, ABYC. This specification's maritime claim is confined to coordination, monitoring, and non-regulated wetting; any function touching vessel safety certification or fixed shipboard fire suppression is deferred to those authorities by reference. Shipboard integration, where undertaken, presents the coordination and monitoring layer to a vessel's existing certified systems and does not substitute for them. As on land, the equipment stays deliberately in the open category and out of the regulated-suppression listing regime.

## Class 3 — Tactical

Transportable, prepositionable node for forestry, National Guard, and emergency management deployment alongside existing pump-and-hose structure protection practice. Zero contact with agency pump equipment required.

- **Enclosure:** ruggedized transit case (Pelican class); identical internals and firmware
- **Power, primary:** battery/solar — portable power station of an open commodity class (1–2 kWh, solar input, user-replaceable as a unit) or internal LiFePO4 with charge controller; solar panel in kit
- **Power, auxiliary — tool-battery dock:** the unit provides a removable-battery dock accepting a major open tool-platform pack (DeWalt 20V/FlexVolt, Milwaukee M18, Makita LXT class), **selected at order time to match the operating department's existing battery fleet** — the same platform already powering the department's battery extrication tools and charging banks. 18–20V pack through DC-DC boost to the 24V bus; one 5Ah pack sustains controls and radio for a full shift; hot-swap sustains indefinitely. Roles: field power without solar, emergency runtime extension, commissioning power. *Equipment requiring proprietary single-source battery systems is expressly non-compliant with this interface.*
- **User charging (interior):** the case interface is the open lid; charging provisions live inside it, unsealed, where the operator already has access to everything. One or more interior USB Power Delivery outlets (commodity 12/24V-input PD modules, minimum 45W each) on the panel plate, powered from the internal supply. **Outlets deliver rated power whenever energy is physically present, at operator discretion; automated policy may inform — status indication, energy events on the mesh — but shall not refuse.** How to spend the node's energy is an operator's judgment. A charging cable of the prevailing consumer device interface ships in the case and is **treated as a consumable** — inventoried and replaced at warehouse maintenance intervals, so a long-prepositioned case opens with a cable matching the devices of the year it opens. An inductive charging pad of the prevailing wireless standard is a **standalone module** — a discrete unit mounted in the lid or panel recess, powered from an interior outlet, added, removed, or upgraded independently of all other equipment, because wireless charging standards evolve faster than the case does. Justification for the procurement file: field personnel operate on battery-powered personal and issued devices; a protected charging point on prepositioned equipment is an operational capability, not an amenity. The local service connection is direct Ethernet or USB at the controller, inside the case.
- **Connectors:** high-exposure tier (M12/Deutsch class); GNSS pattern 3
- **Internal networking:** DIN rail unmanaged industrial Ethernet switch per Class 2, standard equipment; agency-furnished radios and backhaul terminals plug into switch ports per the export-control boundary
- **Radio:** LoRa for dense sites; tactical MANET tier (Silvus/Wave Relay/TSM class) for range, agency integration, and detection payloads
- **Tactical MANET radio — hardware reference (current, will change):** the radio class is defined by capability, never by vendor (see §Radio-Network Classes); the units below are the best current-generation reference for the Tactical case and are expected to change as the field advances. As of this revision the reference radios are the **Persistent Systems MPU5 (Wave Relay)** and the **Silvus StreamCaster 4200/4400 series**. Their published electrical envelopes drive three integration requirements:
  - *Bus compatibility (validated):* the MPU5 accepts 8–28 VDC and the StreamCaster 9–36 VDC (9–50 VDC on the

4400E). The node's 24 VDC bus lands mid-band for both; a reference radio feeds directly from the bus with no additional converter. The 24 V bus voltage is confirmed correct for this radio class — a lower bus (12 V) would crowd the MPU5 ceiling under vehicle-power transients and double the feed current; a higher bus (48 V) would exceed the MPU5 input.

  - *Power budget (sizing driver):* these radios are a different load class than the residential node. Peak draw is roughly **40 W (MPU5, transmit)** to **65 W (StreamCaster, high-power mode)** for the radio alone. The Tactical supply and battery shall be sized for the reference radio's transmit peak **in addition to** controller, valve, and sensor loads — budget on the order of 100 W-plus continuous headroom, not the residential node's modest control-only budget. A Tactical case built on residential power sizing will brown out the radio on transmit; this is the single most common sizing error for the class.
  - *Feed and interface:* radio power runs on conductor gauge and connectors rated for transmit current (≈2.7 A at 24 V for a 65 W radio), not signal-wiring gauge. Both radios present data over standard Ethernet (StreamCaster adds RS-232 and uses a break-out/ODU cable carrying power, Ethernet, and serial on one connector — allow a gland for it). Both can *source* low-voltage power outward (the MPU5 supplies 5 V on its side connectors); the node shall not contend with the radio over powering shared accessories such as headsets. Integration remains at the Ethernet layer per the export-control boundary; the radio is operator- or agency-furnished.

- **Blue force tracking:** Cursor-on-Target publication to TAK infrastructure per Class 2. A crew working a division sees the node as a fixed rally point that can call for help.
- **Satellite emergency module (optional module; expected Tactical configuration):** dual-constellation satellite SOS — Iridium-network (inReach/RockBLOCK modem class) and Globalstar-network (SPOT modem class) — both fitted, for independent orbits and failure modes. Emergency button press transmits distress with GNSS position by four paths simultaneously: mesh distress event, both satellite constellations, and gateway backhaul where present. Satellite services activate under the operator's existing service arrangements; for government operators these are typically enterprise or interagency agreements rather than per-device subscriptions. Per Charter Clause 2, the fire defense function and the mesh distress event never depend on satellite service; the module is a life-safety enhancement riding above it.
- **Hydraulic/engine sensing (non-invasive):**
  - *Inline sensor spool:* short hose section, 1.5" NH forestry couplings, carrying pressure transducer and flow meter; clamps into the discharge lay in under a minute; pressure-loss signatures identify burst or cut lays immediately
  - *Acoustic engine sensing:* commodity electret/MEMS microphone element on an analog input; portable fire pumps run ≥100 dBA; running/stopped by amplitude threshold, RPM class, load state, and fuel-starvation onset by onboard exhaust-note analysis. **Derived machinery state events only; audio is non-transportable per the message enumeration.** Fuel forecasting by acoustic burn-rate inference against known fuel quantity; fuel float optional.
  - No baseline sensor touches, mounts to, or modifies the pump
- **Duty-cycled operation:** zone valves in the distribution lays let the sequencer time-share pump runtime across coverage points; fuel consumption drops by the duty cycle; a single fuel load extends across the ember-cast window instead of running to dry
- **Pump control interface (published ahead of industry support):** 12V nominal control convention compatible with existing portable-pump control panel and battery-kit practice; start/stop/state via the pump relay and enumerated pump-state categories. Current-generation portable pumps (MARK-3 Watson Edition class) are pull-start; baseline relationship is sense-only. The interface definition is procurement language: agencies specifying coordination-interface pump control in tenders provide the demand signal for next-generation platforms.
- **Future electric pump configuration:** battery-electric portable pumps integrate under the same interface; no starter circuit; battery maintained by the node. Full attack-flow electric pumping exceeds practical battery capacity at current energy density; duty-cycled sprinkler protection — this specification's exact use case — is what makes battery-electric portable pumping viable at all.
- **Weight and carry:** the Tactical node case, as shipped ready to commission, **shall not exceed 40 lb (18 kg)**. Kit components — power station, solar panel, gateway terminal — are separately carried items, each individually under the same limit. The limit exists so that one person can move any element of the kit over broken ground; it is a deployment requirement, not a packaging preference. It also forces the correct power architecture: the case carries only enough internal battery to commission and ride through swaps; the power station travels separately. Kit items are designed to **pair for carry** (balanced loads carry better than single loads of half the weight), and every case carries handles sized and placed for a two-person lift as well as one. The design floor is not the professional firefighter; it is the untrained volunteer who showed up to help.
- **Joint inventory doctrine:** Tactical kits are managed as signed inventories per established deployable-cache practice. Issue and return follow the open-lid ritual: open the case, verify contents against the card, self-test, sign,

move.
    - *Inventory card:* laminated, mounted in the lid, itemized to ninety-second verification depth — node, power station, panel, cables (counted), sensor spool, antenna, adapters, charging cable (consumable), batteries with date. The card, the case contents, and the pattern-library kit manifest are one document with three lives.
    - *Inventory self-test:* the guided-setup system's first mode. On internal battery, without water or antenna mast, the unit walks its own card audibly: radio check, GNSS fix, valve cycle, battery voltage spoken, distress-path self-test without transmission. The event — who signed, when, what was present, what the self-test reported — is logged to the cryptographic chain, extending tamper-evident custody from operation to accountability. Annual warehouse shelf checks use the same mode, by any technician, logged to the same chain.
- **Prepositioning:** palletized warehouse storage, drop deployment. **Prepositioned units shall require no external activation, registration, provisioning service, or connectivity to enter service.** Combined with voice-guided setup, a unit requires no manual to enter service. Warehouse maintenance: battery inspection/swap and charging-cable replacement on standard storage intervals, commodity parts, any technician, verified by inventory self-test.

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## Coverage Engineering and Wetting Schedule (All Classes)

### Home Ignition Zone Framing

Coverage design follows the home ignition zone (HIZ) framework established in federal fire science: whether a structure survives is determined by the structure and its immediate surroundings — the receiving sites where embers land — not by the flame front. Static mitigation (defensible space, home hardening) prepares the ignition zone; this system provides the layer static mitigation cannot: **active ignition zone maintenance during the ember-cast window.** Coverage placement shall prioritize receiving sites — gutters, roof valleys, wall lines, fence lines, and adjacent combustible accumulations — over area wetting.

### Receiving Surface Wetting Classes

Each zone shall be assigned a wetting class at commissioning. Wetting classes anchor to the National Fire Danger Rating System (NFDRS) dead-fuel timelag convention, the classification the fire service already operates by.

- **Vegetation zones** (perimeter grass, mulch, shrub lines) inherit NFDRS classes directly. Fine fuels are 1-hour class and re-dry toward hazard rapidly under fire weather; these zones receive the shortest re-wet cycles and full weather sensitivity.
- **Structure zones** are classed by their effective fuel. Untreated wood shake behaves as coarse dead fuel. Class A rated surfaces (asphalt shingle, metal, tile — per ASTM E108/UL 790) are themselves ignition-resistant; **their effective fuel is the accumulated debris** — gutter litter, needle accumulation in valleys — which is 1-hour class material. A rated roof zone therefore still cycles; its schedule follows its debris, not its shingles.

**Adopt-by-reference clause:** wetting schedules shall follow the current published methodology of the national fire danger rating system; revisions to that methodology are adopted by reference and do not constitute revisions to this specification.

### The Cycle Equation

The sequencer's re-wet interval is the solved output of the drying model, not a fixed timer. For each zone: the application pass raises surface moisture above the protective threshold; the NFDRS drying equation (first-order, driven by temperature, relative humidity, and wind) gives the time for that moisture to decay back toward hazard; that interval is the zone's maximum cycle time. Duty cycle is therefore computed continuously as drying time against wetting time. Hot, windy, low-humidity conditions tighten the rotation automatically; recovery conditions relax it and extend supply endurance. Duty-cycled coverage is not a compromise on continuous wetting; it is the solved form of it.

### Weather Sources

The drying model runs on every node; its inputs are a layered service with interchangeable sources, best available first:

1. **Local sensor** — Davis/Kestrel-class station on the node or a domain neighbor; truest, works isolated
2. **Domain-shared weather** — one instrumented node serves the coordination domain via the existing environmental telemetry category

3. **Public feed via gateway** — RAWS, NWS, or MesoWest-class aggregator data injected as domain weather by any node with backhaul; example source classes, not dependencies
4. **Conservative default** — with no weather from any source, the node assumes fire-weather drying rates and cycles accordingly. Isolated nodes err wet, which is the correct failure direction.

Weather sources are an enhancement, never a dependency, per Charter Clause 2.

### Application Rates and Head Selection

Coverage distribution adopts by reference the application rates, overlap, and placement practice of published wildland structure-protection sprinkler guidance, to its current editions — the national cache sprinkler-kit standards and the structure-protection field literature, alongside Canadian SPU and FireSmart-class practice. The forestry-standard brass impact sprinkler (25PJ class) is the reference head for roof and broad coverage; low-flow plastic impacts and micro-heads are the reference classes for thin supplies and spot targets. The **Supply & Coverage Planning Table** (OSF-2, Appendix A) provides conservative planning figures for supply runs, head classes, spacing, and per-zone head counts; the unit's commissioning self-calibration measures actual supply and governs over all planning figures.

**Head placement — damage prevention is a design requirement.** Published structure-protection guidance is emphatic on a point that is easy to get wrong: the objective is to wet the receptive fuels adjacent to the structure and raise humidity around it, not to wash the building. A dwelling's roof and wall assemblies are built to shed water falling from the sky; an impact head drives water horizontally with force, past siding and into the wall cavity. The resulting damage may be immediately visible on an exposed assembly, or may remain hidden and surface months later as mould behind interior finishes. Installations following this specification shall observe the following placement practice:

- **Prefer a settable-arc head over a full-circle head** wherever a head is placed near a wall. The arc is set to throw away from the structure, permitting close-in placement without water striking the building. Full-circle heads are permitted; they require more standoff or an angled placement.
- **With a full-circle head near a wall, place close-in and angled** — staked leaning toward the value at risk, approximately three to four feet off a corner — so the pattern dampens adjacent ground cover without saturating siding.
- **Roof wetting is achieved by lofting, not by driving.** Where the roof plane is the target, water is thrown up onto the deck rather than aimed at the surface, avoiding lifted shingles or shakes and keeping the stream off the walls below.
- **Overlap adjacent patterns.** Spacing follows the Planning Table: spacing equal to the effective radius as the baseline, derated in wind. Spacing at the full diameter of throw leaves a dry gap between heads and does not meet the coverage intent.
- **Design pressure for structure-protection heads is 50 psi.** Below approximately 40 psi the effective radius collapses and any spacing derived at design pressure no longer overlaps. Commissioning self-calibration measures delivered pressure and governs.
- **Feed heads from a trunk with laterals** rather than one long straight run of small-diameter hose, so the last head is not starved relative to the first.

### Spacing and Participation

At suburban structure spacing (separations under approximately 30 feet), structure-to-structure ignition dominates the loss mechanism, and adjacent-structure protection becomes the controlling variable: a participant's survival is partly a function of parcels they do not control. Published multi-fire analyses confirm that structure spacing is a critical loss factor and that community-scale coordinated mitigation is what maximizes effectiveness. At this density, participation density matters more than individual system quality. This is the coordination protocol's founding case restated as fire science.

### Undefended-Parcel Deployment

Coverage may extend beyond ownership. A vacant lot, unoccupied house, or unmitigated parcel between participants may be covered from participants' own supplies: a long supply run (3/4 in. hose or better strongly preferred over distance — see the Planning Table friction figures) feeding one to two low-flow heads placed at the parcel's receiving sites, on its own zone. Self-calibration accommodates the reduced supply automatically; the sequencer schedules the thin zone on its wetting class like any other. The protection delivered is modest by design and sufficient by intent: the gap in the neighborhood's ignition zone is the neighborhood's problem, and this is the tool shaped like the gap.

### Coverage Target: Structure or Terrain

A zone's coverage target may be a structure's receiving surfaces or a piece of ground the fire must cross. The hydraulic and wetting-class treatment is identical — application rate against fuel type under live weather — whether the receiving surface is a roof, a gutter, a fence line, or the surface fuel in a drainage bottom. The terrain-target case is developed as doctrine in the specification's expeditionary section (Terrain-Anchored Deployment and Non-Destructive Suppression); at the annex level the relevant fact is that the coverage-engineering math does not distinguish between wetting a structure and wetting a corridor, and the same node, valves, and sequencing serve both. A structure that cannot be defended at its own walls may be defended by wetting the terrain upstream of it.

### Worked Example — Post-War Ranch Archetype

Representative case: single-story ranch, approximately 1,100 sq ft, Class A asphalt roof, municipal supply at 50 psi, 18-foot separation to neighboring structure with wood fence in the gap, mature deciduous trees over the roof line.

- **Zone 1 (roof/gutter):** one 25PJ-class brass impact at the ridge or two small plastic impacts at the eave line; wetting class: 1-hour (debris-governed)
- **Zone 2 (perimeter/vegetation):** low-flow heads on the grass ring and foundation line; wetting class: 1-hour
- **Zone 3 (exposure):** heads directed at the shared fence line and the gap face of the structure; wetting class: coarse fuel (fence) — this zone exists because of the 18-foot spacing, per the Spacing and Participation section
- Total draw per active zone: 3–6 GPM — within a single hose bib. The unit self-calibrates, sequences the three zones on live drying intervals, and participates in domain coordination.

## Expansions (Orthogonal to Class)

**Pump and reservoir expansion** — any class. Cistern per main specification (minimum 2,500 gal cistern-primary; buffer tanks on reliable municipal supply). Pump: self-priming, fuel-drive preferred, minimum 40 GPM at 80 psi, remote start via controller relay (Honda WX class, trash pump configurations, industrial centrifugal electric). Valves move external: ISO 5211 pad bodies with separable 24VDC actuators sized to bore. Sensing additions: submersible 4–20mA reservoir level; pulse-output flow meter on discharge (GPI, Seametrics class). Sensor interfaces at expansion tier: 4–20mA loop **or** Modbus TCP, installer's selection, both native to the controller.

**Zone expansion modules** — any class. Same valve family, small NEMA enclosure, 4-wire daisy-chain bus.

**Gateway kit** — one per coordination domain, single pallet line item: satellite terminal, power station, solar panel. One gateway gives an entire deployed domain — a valley of nodes — external visibility and ledger publication through standard mesh routing.

**Detection payload** — Class 2/3, government tier: thermal camera on tripod. Detection *events* (occurrence, bearing class, confidence class) ride the coordination protocol; imagery, where an agency requires it, rides agency radio infrastructure as agency traffic outside this specification's enumeration. Aerial detection platforms (UAS overwatch) integrate as detection payloads under the same pattern: detection events on the coordination protocol, imagery on agency transport. Adjacent agency security/surveillance assets coexist on the same network without being absorbed by this specification.

**Automated fire lookout configuration** — the named Class 3 composite: Tactical node + pump and hose kit with sensor spool and zone valves + weather station (Davis/Kestrel class; wired/serial preferred; vendor short-range wireless is a local sensor link terminating at the node per Clause 6; local wind feeds wet-cycle scheduling and gives incident command spot weather at the point of greatest forecast uncertainty) + detection payload at government tier. An automated fire lookout that defends its own site — resolving the historical weakness of the staffed lookout, which could see fire but not survive it.

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## Valve Scaling Table

The control object is constant. Only body and actuator scale.

| Bore | Body type | Actuator class (typ. torque) | Application |
|---|---|---|---|
| 3/4" – 1" | Full-port brass ball, motorized or ISO 5211 | Integrated / ≤ 20 Nm | Class 1 zones, Class 3 lays |
| 1.25" – 2" | Full-port brass/SS ball, ISO 5211 | 20–50 Nm, 24VDC quarter-turn | Expansion zones, light commercial |

| | | | |
|---|---|---|---|
| 2.5” – 4” | Lug or wafer butterfly, ISO 5211 | 50–150 Nm | Commercial, small municipal |
| 6” – 8” | Lug butterfly, ISO 5211 | 150–400 Nm | Federal facility, municipal main class |

All actuators: 24VDC quarter-turn electric, open/close, dual limit-switch feedback. Modulating (4–20mA positioner) actuators are the drop-in upgrade at any bore. **Spring-return (fail-closed) baseline**: loss of power drives the valve closed without controller action, which is what makes the disconnect a true hardware kill. Fail-in-place actuators are a permitted per-zone variant where a design or code requirement calls for held position, but a node built entirely on fail-in-place actuators does not satisfy the power-loss-closes property this specification relies on.

## Class 1 Reference Bill of Materials

| Item | Qty | Example products | Est. cost |
|---|---|---|---|
| NEMA 4X enclosure 16×20×8 | 1 | Hoffman, Attabox, Saginaw | $150–300 |
| Motorized 3/4” ball valve, 24VDC, feedback | 3 | CR-series class | $120–210 |
| Inlet check + isolation valves, gallery fittings | 1 set | Commodity brass, NPT | $60–120 |
| Pressure transducer 0–200 psi 4–20mA | 1 | Commodity industrial | $40–80 |
| ESP32 industrial PLC | 1 | Industrial Shields ESP32 PLC 21, Norvi ENET | $150–250 |
| LoRa mesh node + antenna + N bulkhead | 1 | RAK4631, T-Beam | $60–120 |
| GNSS receiver + active antenna + SMA bulkhead | 1 | u-blox class + commodity puck | $30–60 |
| DIN 24VDC supply w/ DC-UPS | 1 | Mean Well DRC-60 | $70–120 |
| Batteries, 12V 7–9Ah SLA | 2 | Commodity panel batteries | $35–50 |
| DIN rail, terminals, wiring, cord, strain relief | 1 set | Commodity | $60–100 |
| GHT adapters, drain valve, bulkhead fittings | 1 set | Commodity brass | $40–70 |
| **Class 1 unit total (parts)** | | | **$815–1,480** |

Class 2 adds connector and enclosure upgrades and radio options against the same internals. Class 3 pricing is dominated by case, power station, and radio tier selection. Costs representative at July 2026 through standard industrial and alarm-supply channels.

## Compliance Statement

An installation is compliant with this annex when every capability requirement above is met, regardless of component substitution. Named products are examples only. Improvements to this annex through community development are welcomed and integrated into subsequent versions under the same permissive licence as the main specification.

## Accessories — The Smart Water-Appliance Family

The fire service’s manual water-handling layer — hydrant operation, gate valves, wyes and water thieves, manifolds, inline shutoffs — is a small, closed set of appliances in a small, closed set of sizes, all riding the national standard

hose thread (NFPA 1963)s already universal on hydrants and apparatus. This section discloses their remotely-commanded siblings. The thread standard did the interoperability work a century ago; the smart family inherits it — no new coupling, no new size chart, no modification to any hose, hydrant, or apparatus.

**Doctrine: sparse by design.** The smart family *follows* the manual system; it does not replace it. The manual layer remains the backbone. Smart appliances deploy only at selected key junctions where remote command pays — the hydrant feeding an operation, the wye at a critical branch, the manifold at a buffer site — chosen by the incident commander or planner, not by default. Every smart appliance retains full manual operation when unpowered; the smart version is a powered hand, never a replacement for one. Placement advice may be drawn from hydraulic calculation and from the system's own accumulated ledger data (an AI or analyst trained on the domain's hydraulics and event history can recommend which junctions earn actuation), but placement is always a human decision.

**Remote hydrant valve actuator.** A clamp-on, non-invasive rotary tool engaging the hydrant operating nut — no modification to the hydrant — electrically driven, remotely commanded through the standard node interface with a two-verb vocabulary (OPEN / CLOSE). The device drives the valve to thread-bottom in the commanded direction and self-terminates; detection of travel completion is left to the implementation (torque-aware self-terminating drive is established art in commercial impact tools). Operates only under the authority of the agency owning the hydrant. Tactical value includes closing hydrants behind a withdrawing crew without a person standing at the wrench.

**Smart appliances (gates, wyes, thieves, manifolds, shutoffs).** Remotely commanded versions of the standard manual appliances in the standard hose-thread sizes. Each is electrically actuated, node-standard in communications and command security, commanded through the closed vocabulary (OPEN / CLOSE per port; throttled and position-reporting variants contemplated within this disclosure), retains manual operation when unpowered, and uses the existing thread standard as its entire mechanical interface.

**The buffered supply point ("a water tower from tender loads").** A portable-tank drop site — multiple standard portable tanks, a multi-port smart manifold in a transit case, and an observation node (camera and/or level sensing) watching tank levels — becomes managed buffered storage: discrete tender deliveries accumulate into standing capacity, delivered as continuous managed supply, with refill and tank-swap triggers driven by observed actuals rather than guesses. The shuttle loop gains a gauge; supply state becomes visible on the command surface and failures of the D-7 kind (a supply point silently dying) become visible hours before they bite. This is the strongest single argument for cached water: storage plus observation plus a commandable manifold equals a water tower assembled from shuttle loads, anywhere a tender can reach.

**Power and compliance.** Compliant implementations of every appliance in this family accept the system's common battery interface per the family's commodity-battery doctrine. Proprietary-power and alternative drive-mechanism variants are contemplated within this disclosure and are disclosed herein so that they cannot subsequently be patented by others; as system devices they are non-compliant. Vendors are invited to compete on quality — actuation, sealing, ergonomics, runtime — on shared threads, shared battery, and the shared closed vocabulary.

**The node-waypoint patrol (aerial observation without towers).** Every node publishes its position; the domain census is therefore also a UAS patrol route — node coordinates plus an altitude offset generate a waypoint mission automatically, and a drone flying the circuit captures a snapshot of each node, tank site, and appliance per pass. Visual level-check by flyby substitutes for a fixed observation node wherever a camera tower cannot be sourced, spared, or justified; automated waypoint inspection from asset positions is established commercial art (docked autonomous patrol systems), and this disclosure claims the integration: domain census as auto-generated patrol mission, flyby observation as a peer of fixed observation on the command surface. Airspace conduct is adopted by reference: on incidents, per NWCG PMS 515 — no launch without positive contact with aerial supervision, operation within the assigned altitude block inside the Fire Traffic Area, manned aviation above; outside incidents, per FAA Part 107 and successor beyond-visual-line-of-sight rules as they mature. The patrol serves observation only; it commands nothing, per the standard security model.

**Thermal-signature verification (protection state as a map).** A wetted surface evaporatively cools; a dry or failed surface does not. The contrast is stark in the thermal band, so a single thermal pass over a coordination domain renders whole-domain protection state as a human-readable image: cool structures are being wetted, hot structures are not. This is external, physics-based ground truth — it measures actual water on the actual surface rather than asking any node to report its own state, and therefore catches the failure modes self-reporting cannot: a clogged head, a kinked line, a valve that confirmed open and delivered nothing. The thermal read is disclosed as a payload variant of the node-waypoint patrol above: the same auto-generated circuit, flown with a thermal camera, produces a periodic protection-state map at range without any crew setting foot on a property.

It stacks with the signed operational ledger to close the last gap in the record. The ledger attests what was *commanded*; the thermal pass attests what physically *happened*. Together they are command-plus-confirmation, and the pairing is what allows an after-event report to survive a reader who does not trust the operator. Timestamps and zone identities in the ledger are recorded so that an external thermal observation can be correlated against them directly.

As a maintenance instrument the same pass is an inspection. Flown before a season, or during a commissioning test cycle, the hot roofs are the work order — a whole neighbourhood is verified functional in one overflight rather than one property visit at a time. A configuration in the shared library is proven when the roof reads wet. Observation only; it commands nothing, per the standard security model.

### Disclosure — Utility Right-of-Way Corridor Defense

A coordination domain may be deployed along electric utility rights-of-way as corridor defense: detection nodes at spans and crossings identified by the utility's own vegetation and ignition history, terrain-defense wet line at high-risk crossings and structure-adjacent segments, buffered supply points at substations and access points, and node-waypoint patrol routes generated from the corridor census for vegetation-encroachment and post-event holdover inspection. The domain's signed operational ledger serves as the utility's verifiable mitigation record: after-event reports demonstrating that corridor defense existed, operated, and responded are generated from the ledger rather than from operator narrative. All equipment, message categories, trust-layer behavior, and charter constraints are unchanged; the corridor is a deployment geometry, not a new system class. Utility-furnished transport, where provided, integrates at the network layer per the layered architecture.

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# Appendix A — Common Parts Catalog

*Reference configuration. Consolidated into this annex; previously issued separately. The parts named here are a reference implementation, not a requirement — any component meeting the capability requirements of this annex is compliant.*

### What this is (and is not)

This is **a** known-good reference configuration, not **the** design. It names one working part per function — biased toward the better version (a MeanWell supply, not a generic; the oversized enclosure, not the tight one) so a first build is likely to work rather than likely to teach the wrong lesson. Every entry is tagged with the *function and requirement* it fills, so a builder reads it as a starting point and substitutes freely against the requirement (per Annex A capability-substitution). The catalog fills in over time: community builds report alternatives, regional availability, and price tiers under each function. Quality-biased for prototype success; substitution-permitted forever after.

**This is a starting place for others to build from.** It is not a bill of materials and it is not prescriptive. The reference part is the one we could verify openly and build with confidence — nothing more.

### Sourcing rule — vendor accessibility tiering

Reference parts are selected in part by **how openly the vendor publishes specs**, because that property serves the builder exactly as it serves the catalog: an openly-specced part is one the next builder can verify and substitute. Vendors are tiered:

- **Tier A (preferred):** public downloadable datasheet PDF or full web spec, no login; **bonus:** downloadable CAD / STEP / DXF stencils for panel layout. *(e.g. MeanWell — predictable per-model spec PDFs; AutomationDirect — per-part Docs & Drawings tab with datasheet + 3D CAD in 60+ formats.)*
- **Tier B (acceptable):** full specs on the website, no login, no CAD.
- **Tier C (bumped down):** specs behind a login, registration wall, or rep-request. Usable only if no Tier A/B part fills the function.

A Tier-A part with a downloadable stencil is worth more to this catalog than a marginally better part behind a login, because the catalog's job is to be *built from*, and you cannot build easily from a spec you cannot pull.

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### The Reference Configuration — Datasheet (skeleton)

*Columns: subsystem function · requirement it must meet · reference part (to fill) · vendor tier · datasheet/stencil status · notes. Empty cells are open slots to fill in over time.*

#### 1. Power

| Function | Requirement | Reference part | Tier | Spec / stencil | Notes |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Primary AC/DC supply | 24 VDC, DIN-rail, UL 508-listed, wide-temp, derated headroom | MeanWell HDR- / NDR- series *(model TBD by load)* | A | PDF ✓ | Oversize for margin; step-shape slim body |
| DC-DC / logic rail | 5 V or 3.3 V from 24 V bus, DIN or board | MeanWell DDR series *(TBD)* | A | PDF ✓ | 4:1 wide input option |
| DC UPS / battery control | Battery-backed 24 V hold, monitored | MeanWell DR-UPS40 class *(TBD)* | A | PDF ✓ | Fills 72-hr autonomy w/ external battery |
| Battery | LiFePO4 or SLA, sized to autonomy target | *(open)* | | | Chemistry TBD per class |
| Solar input / charge (optional) | MPPT, node-scale | *(open)* | | | Tactical/remote tier |
| Surge / inrush protection | Per Part 15 / UL 508A practice | *(open)* | | | |

### 2. Control

| Function | Requirement | Reference part | Tier | Spec / stencil | Notes |
|---|---|---|---|---|---|
| Compute / controller | Runs closed command enum, low power, field-serviceable | *(open — ESP32-class or industrial equiv.)* | | | Board vs. industrial PLC TBD |
| DIN rail | 35 mm TS35 steel/aluminum | AutomationDirect DIN rail | A | PDF + CAD ✓ | Stencil available |
| Terminal blocks | Screw/screwless, 24–3/0 AWG range, UL | AutomationDirect DINnector series | A | PDF + CAD ✓ | Ground blocks green-yellow |
| Relays / valve drivers | Drive motorized valve actuators, DIN, isolated | *(open)* | | | Dry-contact or PLC-card drive per actuator |
| Fusing / protection | Per subsystem | *(open)* | | | |

### 3. Comms

| Function | Requirement | Reference part | Tier | Spec / stencil | Notes |
|---|---|---|---|---|---|
| Mesh radio (baseline) | LoRa class, license-free, neighborhood range | *(open — Meshtastic-compatible module)* | | | Field-proven range basis |
| Tactical MANET (option) | Self-forming mesh, IP, MIL-STD-810 class | Agency-furnished (Silvus/Wave Relay/TSM class) | — | — | Operator-furnished; not catalog-sourced |
| Gateway / backhaul | Satellite or cellular uplink per domain | *(open)* | | | One per coordination domain |
| Antenna | Band-matched, weatherized | *(open)* | | | |

### 4. Sensing

| Function | Requirement | Reference part | Tier | Spec / stencil | Notes |
|---|---|---|---|---|---|
| Water pressure | Inlet pressure, | *(open)* | | | Drives supply-sag |

| | | | | | |
|---|---|---|---|---|---|
| | analog/digital, potable-safe | | | | yield behavior |
| Flow (optional) | Zone flow verification | *(open)* | | | |
| Temperature / RH | Local weather, drying model input | *(open)* | | | |
| Wind (weather-station config) | Anemometer for forward node | *(open)* | | | Expendable forward posting |
| Pump acoustic (non-contact) | Engine-run detection, zero mod to agency gear | *(open)* | | | Acoustic DSP |
| Flame / heat detection (option) | Thermal, per detection payload | *(open)* | | | Feeds CoT `d-` branch |

### 5. Hydraulics

| Function | Requirement | Reference part | Tier | Spec / stencil | Notes |
|---|---|---|---|---|---|
| Zone valves | Standalone motorized full-port brass ball valve, 24 V, ISO 5211 pad, position feedback | *(open — industrial actuated-valve class)* | | | Reference valve; brass body suits the 4X wet section. Solenoid irrigation valve is a permitted budget variant where position feedback is deliberately waived |
| Sprinkler heads | Impact / rotor, published precip rate | *(open — 25PJ-class impact)* | | | Reference precip from vendor tables |
| Manifold / fittings | Poly / brass, standard thread | *(open)* | | | |
| Pump (tactical/terrain) | Portable, draft-capable, gph per water plan | *(open — WATERAX class)* | | | Forestry-standard |
| Supply line | 3/4”+ hose or pipe, friction per planning table | *(open)* | | | See OSF-2 Appendix A friction figures |
| Strainer / suction | Draft from static or open water | *(open)* | | | Millpond / tank / drum |

### 6. Enclosure

| Function | Requirement | Reference part | Tier | Spec / stencil | Notes |
|---|---|---|---|---|---|
| Residential enclosure | NEMA-rated, oversized for margin, UV-stable | *(open — Hubbell-Wiegmann / AutomationDirect class)* | A | PDF + CAD ✓ | Oversize deliberately |
| Tactical case | Transit-rated, foam-fit, ≤40 lb ready | *(open — Pelican-class)* | | | Prepositioning / drop |
| Marine-hardened (Class 2) | 316 SS wetted, sealed connectors | *(open)* | | | Per maritime use cases |
| Cable glands / | Sealed, | *(open)* | | | |

| | | | | | |
|---|---|---|---|---|---|
| entries | strain-relieved | | | | |
| DIN backplate / subpanel | Sized to layout | AutomationDirect subpanel | A | PDF + CAD ✓ | Stencil available |

## How this fills in

1. **Web-planner pass (this document's method):** the panel side is populated from open web configurators and per-part datasheet/CAD downloads (MeanWell, AutomationDirect first). The irrigation side is populated the same way from the sprinkler vendors' public design tools and precipitation-rate tables.
2. **Reference-build pass:** parts are proven in an actual bench/pilot build; anything that flakes is re-specced and noted.
3. **Community pass:** builders report alternatives, regional availability, price tiers, and substitutions under each function; the catalog accretes columns over time.
4. **Sample-hardware (post-attention only):** once the standard has visibility, vendors of Tier-A openly-specced parts are the natural first providers of sample hardware for reference builds — they already publish everything needed to be included, and a reference slot is a placement worth their sample. *No sourcing relationships pre-filing.*

*This catalog is a starting place, not a specification. The reference part is the one we could build with confidence and verify in the open — substitute freely against the stated requirement. It fills in over time; that is the design.*